



Qualcomm Technologies International, Ltd

ADK Tools and debugging

80-CF180-1 rev AA (CS-00405513-TM)
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ADK Tools and Debugging

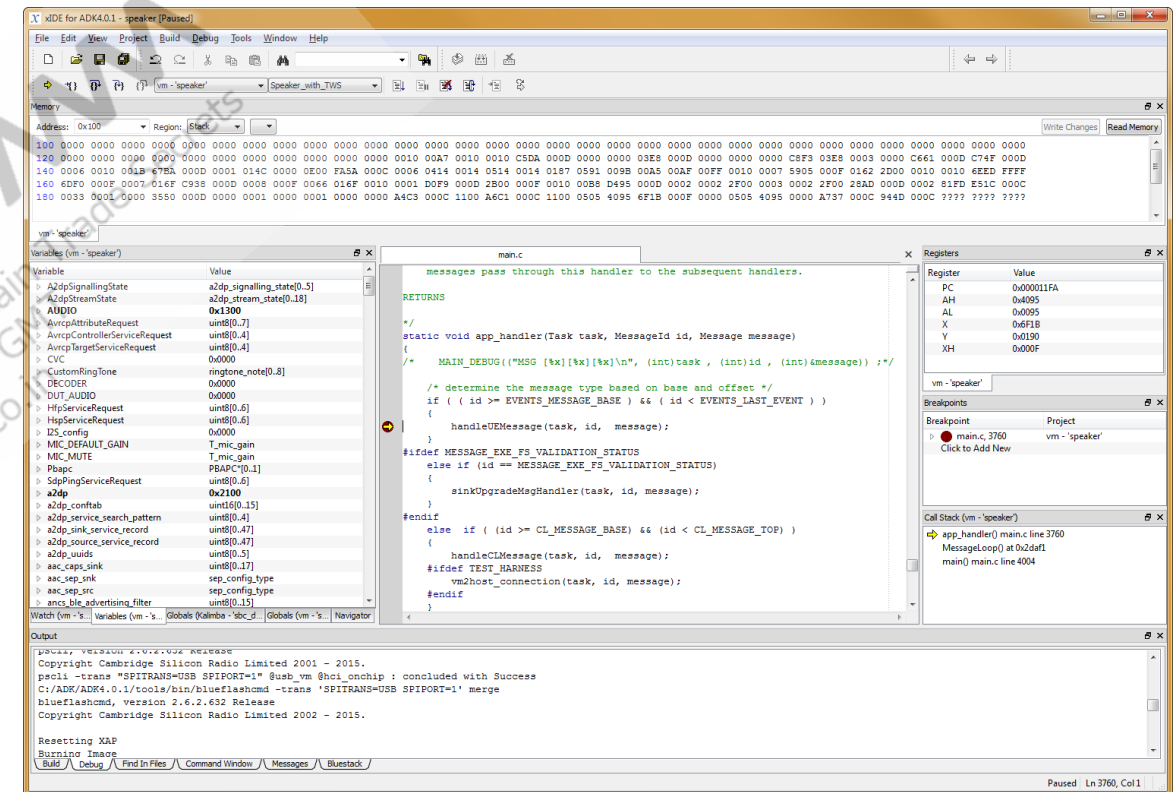
xIDE Overview

xIDE is the Integrated Development Environment for CSR867X and QCC300x devices

Includes the build tool chain (compiler, linker, etc) as well as a debugger

Additional information:

- BlueLab xIDE User Guide (CS-00101500-UG)

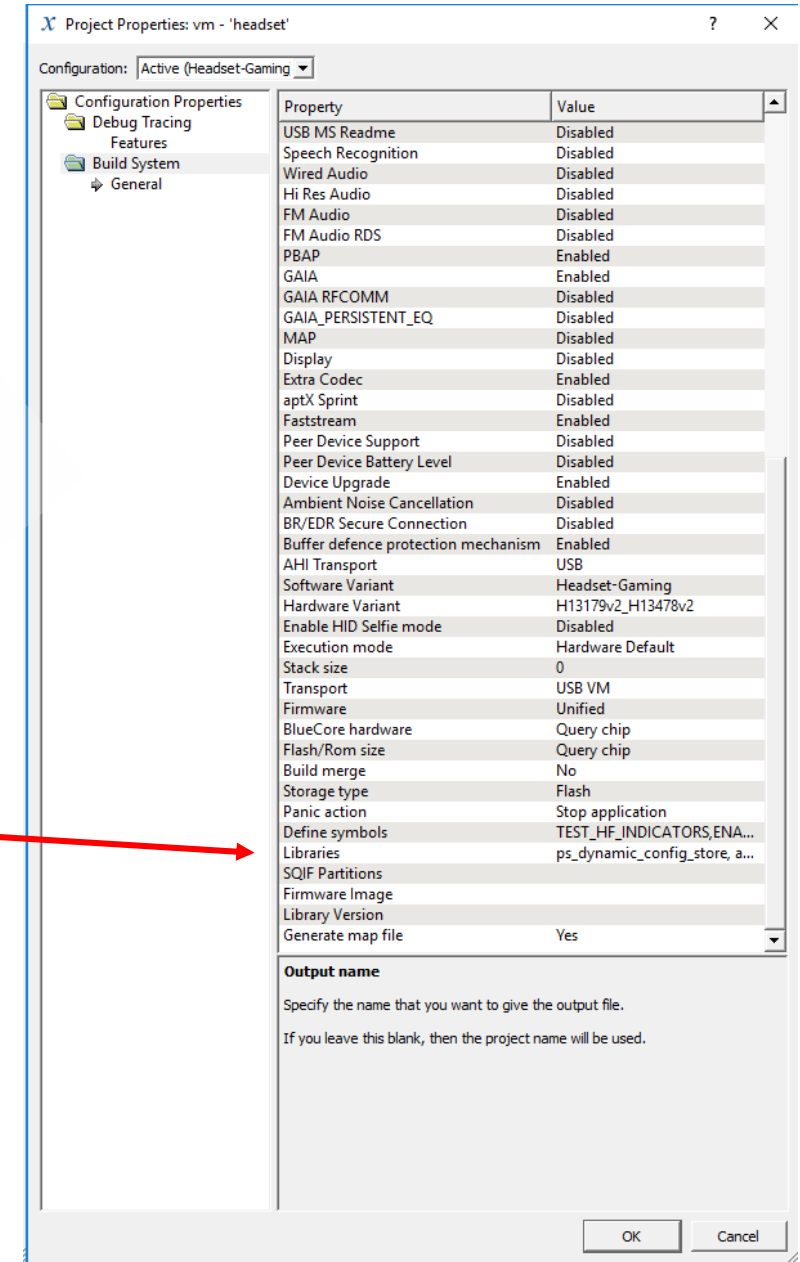


ADK Tools and Debugging

Project Properties

For each of the projects in the workspace, build options are defined and saved as a configuration file (`_vmprops.xml`)

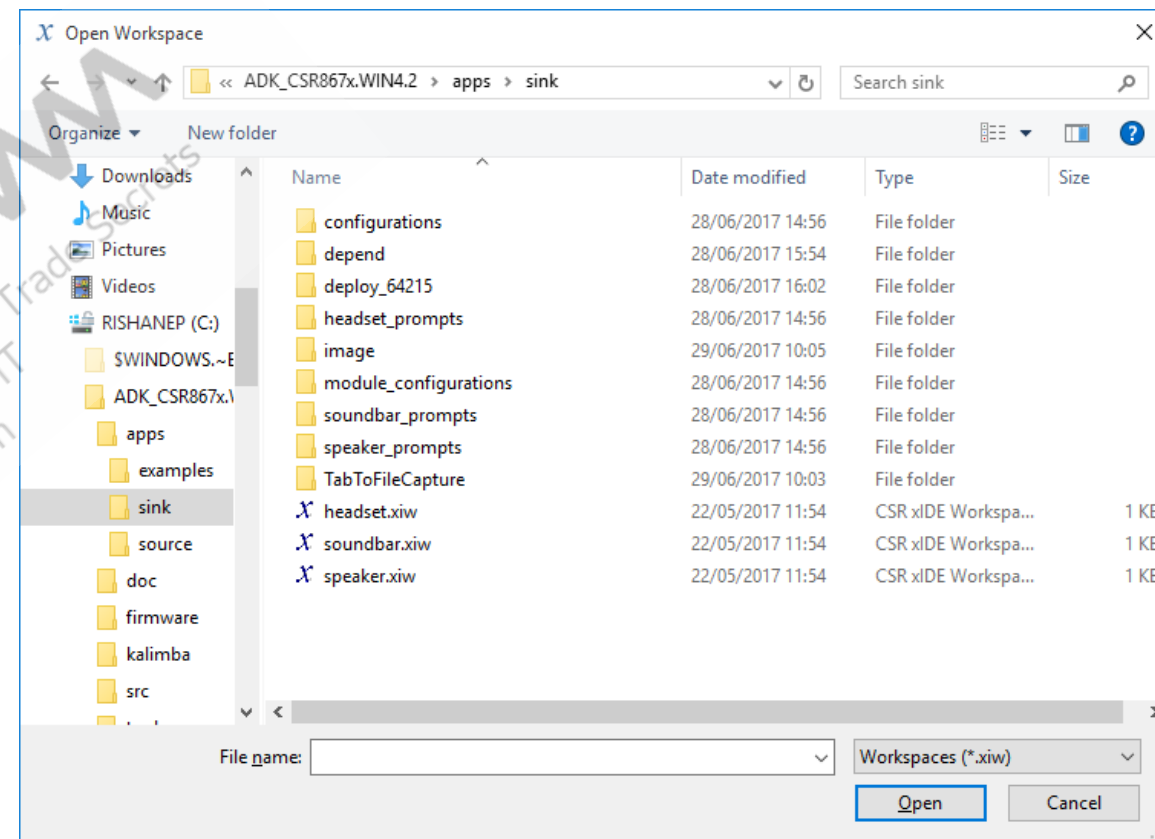
The Libraries defined in the options are linked during the build process



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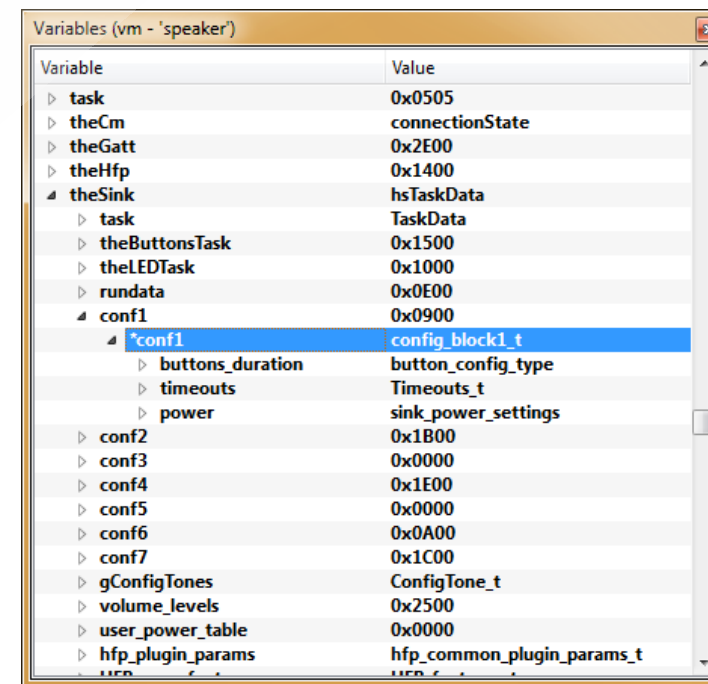
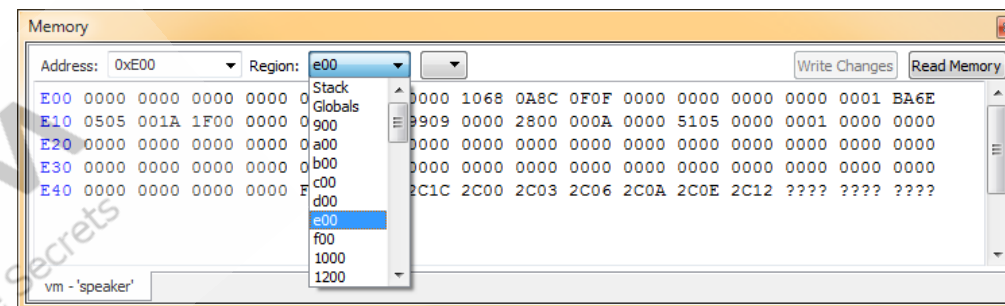
ADK Sink Application in xIDE

- The Sink Application in the ADK includes pre-configured projects to create the following applications:
 - Headset - battery powered headphones
 - Speaker - battery or mains powered speakers
 - Soundbar - mains powered device
 - Only available on CSR867X Premium Flash devices
- The example xIDE workspaces provided for each of the above applications configures the most commonly used options for each of the builds.
- Designed to help customise features by altering the project properties.



ADK Tools and Debugging Views (1)

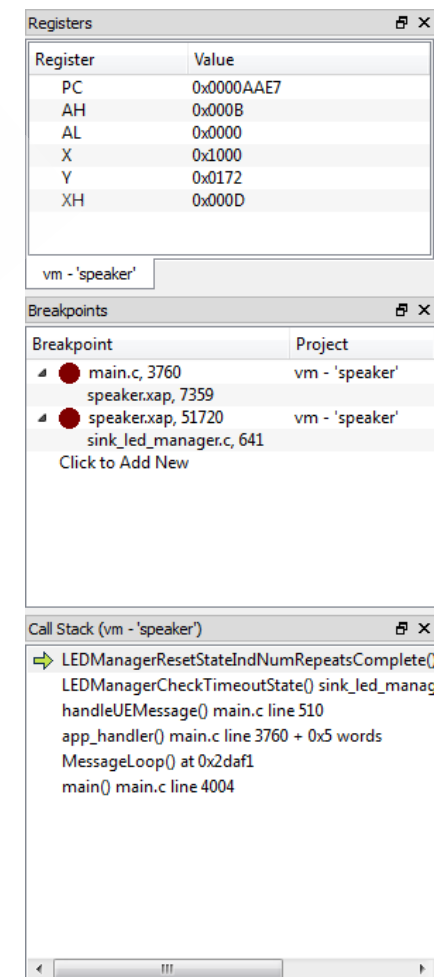
- Memory view: Check the contents of any memory that is allocated to the VM.
- Variables view: Access program variables (Not in real-time).
 - This is especially useful to navigate through complex structures



ADK Tools and Debugging

Views (2)

- Registers view: See the registers content
 - Useful to follow the assembly instructions.
- Breakpoints may be set on C or Assembly instructions
- Call stack view: See which code paths are being taken
 - Useful when debugging functions that are called from multiple places.



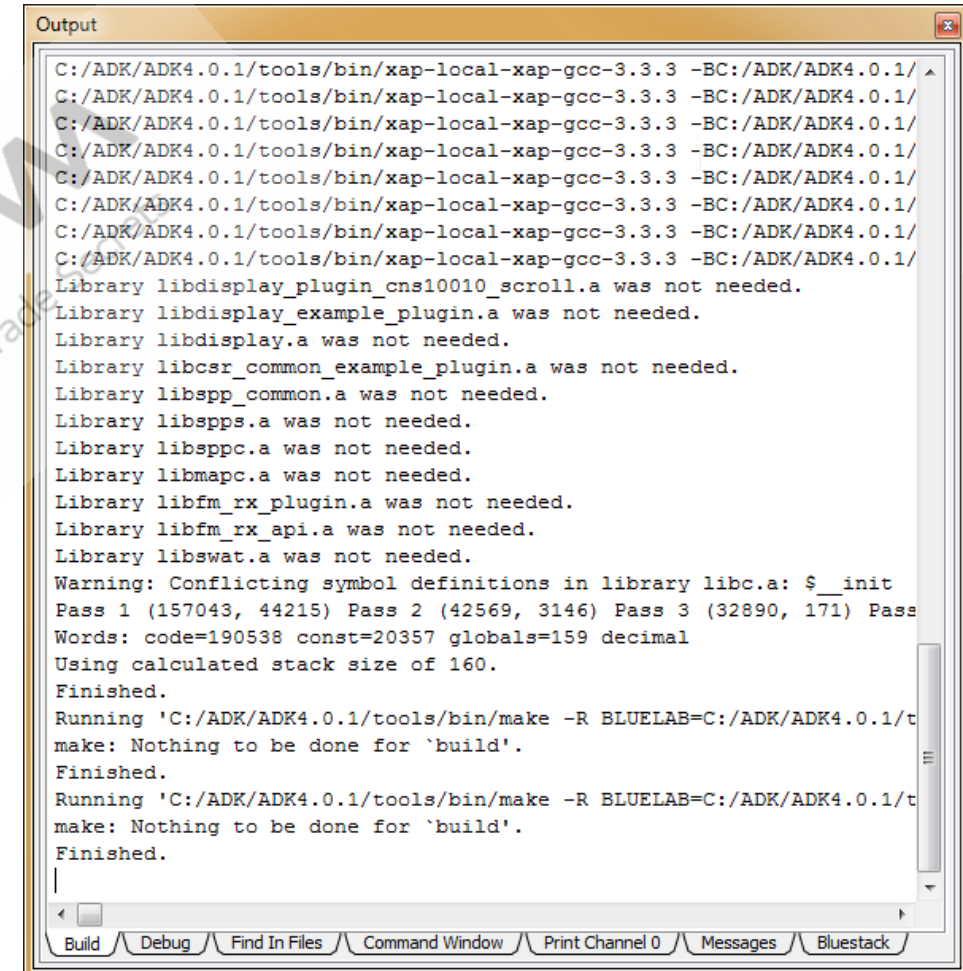
ADK Tools and Debugging

Views (3)

Output view: Displays the following information tabs

- **Build** - Output of the compile and link process for each project
- **Debug** - Output of the image creation process
- **Print Channel 0** - Debug prints from the application running
- **Messages** - Messages sent or received by the VM tasks
- **Bluestack** - Bluestack primitives exchanged with the firmware

The Output view has a limited buffer so if a long output is required, the contents may have to be copied multiple times over the course of the test to assure no data is lost.



The screenshot shows a window titled "Output" with a scrollable text area. The text contains multiple lines of build output, including paths to GCC and BC tools, and a list of libraries that were not needed. At the bottom of the window, there are tabs for "Build", "Debug", "Find In Files", "Command Window", "Print Channel 0", "Messages", and "Bluestack".

```
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
C:/ADK/ADK4.0.1/tools/bin/xap-local-xap-gcc-3.3.3 -BC:/ADK/ADK4.0.1/
Library libdisplay_plugin_cns10010_scroll.a was not needed.
Library libdisplay_example_plugin.a was not needed.
Library libdisplay.a was not needed.
Library libcsr_common_example_plugin.a was not needed.
Library libspc_common.a was not needed.
Library libspcs.a was not needed.
Library libspcpc.a was not needed.
Library libmapc.a was not needed.
Library libfm_rx_plugin.a was not needed.
Library libfm_rx_api.a was not needed.
Library libswat.a was not needed.
Warning: Conflicting symbol definitions in library libc.a: $__init
Pass 1 (157043, 44215) Pass 2 (42569, 3146) Pass 3 (32890, 171) Pass
Words: code=190538 const=20357 globals=159 decimal
Using calculated stack size of 160.
Finished.
Running 'C:/ADK/ADK4.0.1/tools/bin/make -R BLUELAB=C:/ADK/ADK4.0.1/t
make: Nothing to be done for 'build'.
Finished.
Running 'C:/ADK/ADK4.0.1/tools/bin/make -R BLUELAB=C:/ADK/ADK4.0.1/t
make: Nothing to be done for 'build'.
Finished.
```


ADK Tools and debugging

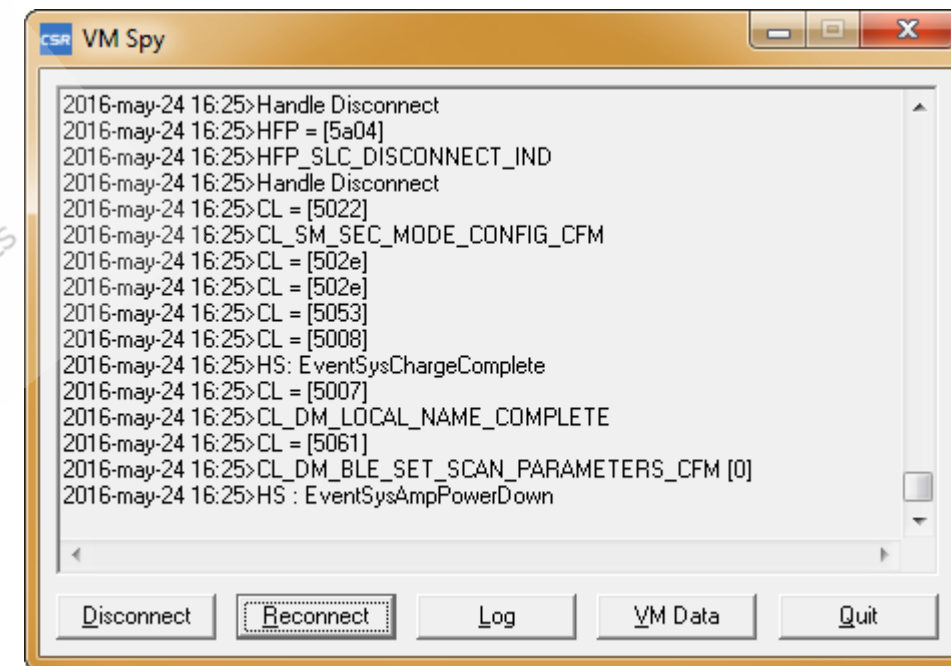
VMSpy

The VMSpy tool allows the user to connect to the VM from outside the xIDE debug mode.

- Can also use it as a simple terminal application to access BCSP channel 13.
- Main usage is usually to display the contents of the *Print Channel* output tab.
- Possibility to write the entire output to a file, especially useful for longer tests

To configure the VM for the tool:

- Set PSKEY Host Interface to *UART link running BCSP*
- Set UART baud rate to 38400
- Disable USB enumerations in Sink config Tool
 - UART and USB cannot be used simultaneously



Additional information on VMSpy:

- VMSpy User Guide (CS-00123515-UG)

Thank you

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arumugaraj@qualcomm.co.in

